

Time Series Forecasting: Session Learning Objectives

After this session, you will

- ① Learn some simple facts about forecasting
- ② Differentiate between *ex post* and *ex ante* forecasting
- ③ Learn what values can be forecast, how far out and with what accuracy.
- ④ Look at overfitting & underfitting and the tradeoff between Bias & Variance
- ⑤ Compare different models for Forecasting in Time Series Analysis:
 - + Naïve models
 - + Linear & Non-Linear Models
 - + Moving Average Smoothing Models
 - + ARIMA & SARIMA Models
- ⑥ See how the above perform for different examples.

Basics: Definitions and Notation

- Many instances when we need to forecast using time series:
 - + Making company hiring decisions 5 years ahead based on long term sales
 - + Making telecomms routing decisions for a few minutes beforehand
 - + and a whole host in between
- Further, predictability of event/ quantity depends on factors e.g.:
 - + how well we understand the factors that contribute to it;
 - + how much data is available;
 - + whether forecasts can affect what we are trying to forecast.

Forecasting: Which Method to Use - Qualitative or Quantitative?

- The forecasting method to use depends on data available:
- If none available, **qualitative forecasting methods** must be used.
 - + these are subjective but not purely guesswork
 - + some well-developed, structured approaches to get good forecasts without historical data.
 - + examples include: informed opinion & judgment, market research etc.
- **Quantitative forecasting** forecasts future data as a function of past data and works when two conditions hold:
 - 1 numerical info about the past is available;
 - 2 we can assume some aspects of past patterns will persist.

Ex post Vs Ex Ante Forecasting

- *ex ante* vs *ex post*

- + *ex ante*

- those made using only info available in *advance*.
 - 'forecasts' are made before the outcomes have occurred
 - evaluated at a later stage when the outcomes are known

- + *ex post*

- those made using later info on the independent variables.
 - forecasts are evaluated against a subset of the originally available data
 - this has been 'retained for in-sample forecasts'

- *ex ante* values are considered as true forecasts;

- *ex post* values are useful for studying forecasting models' behaviour.

Forecasting Models 0: The Naïve Model

- The Simplest quantitative method, where forecast value simply equals previous one, is the **Naïve method**:

$$\hat{y}_{T+H|T} = y_T \quad (2.2)$$

$\hat{y}_{T+H|T}$ is a short-hand for the estimate y_{T+H} based on y_T

- However, generally used only for comparison with the forecasts generated by the better (sophisticated) techniques.
- Can see how the this model derives from the Simple Exponential Smoothing Model
- Recall from SES definition previously, that:
$$\hat{y}_{T+H|T} = \alpha y_T + (1 - \alpha) \hat{y}_{T|T-H}$$
- If we substitute $\alpha = 1$, we get the above expression in eqn (2.2)

Forecasting Models 1: (Linear) Regression

- Prediction on the forecast variable y_t can be made using the regression:

$$\hat{y}_t = \hat{\beta}_0 + \hat{\beta}_1 x_t + \hat{\beta}_2 x_t^2 + \cdots + \hat{\beta}_k x_t^k \quad (2.3)$$

where $\hat{\beta}_i$ s are estimated coefficients, $t = 1 \dots T$.

- Note: the $\hat{y}_t$ values are based on *ex ante* values of $\hat{\beta}_i$
- In its simplest form eqn (2.3) reduces to: $\hat{y}_t = \hat{\beta}_0 + \hat{\beta}_1 x_t$ (see Fig 2.33)

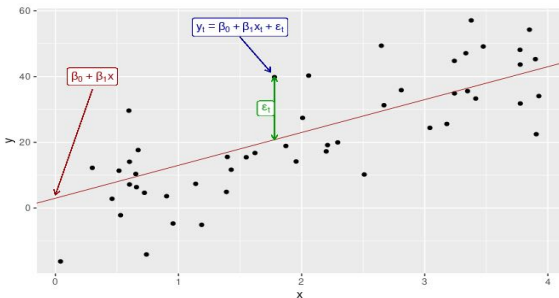


FIGURE 2.33: A Simple Linear Regression Model

Forecasting Models 1: (Linear) Regression (/2)

Forecast with `tslm()` for Australian Beer Series from, 1991-97

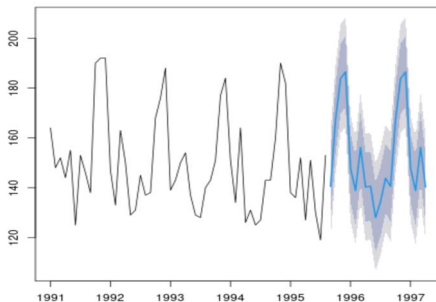


FIGURE 2.34: Australian Beer Production in Megalitres using `tslm` in R

- R's `tslm` is used in Fig 2.34 to fit a linear model
 - `tslm()` creates a simple trend/ seasonal model from given data.
 - Here, beer data has just a seasonality pattern during 1991-97
 - The dark/light shaded region show 80%/95% prediction intervals.

Forecasting Models 1: (Linear) Regression (/3)

Forecast with `tslm()` for Air Passenger Series from, 1948-65

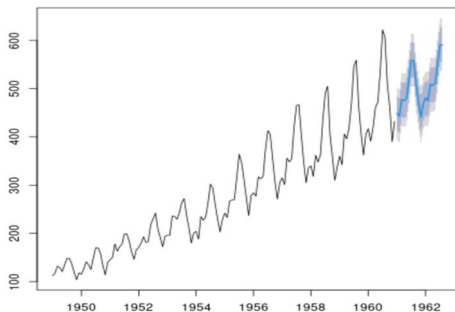


FIGURE 2.35: Air Passenger Data using `tslm()` in R

- Fig 2.35 shows a series with seasonality and trend using `tslm()`
- `tslm()` is a modified version of R's standard linear model `lm()`
- Again, 80%, 95% prediction intervals shown in dark, light regions.

Forecasting Models 1: Regression

- Generally, best to try linear first as it is simpler to interpret and fit.

Linear Regression

- Easier to fit and explain.
- Restricted in the shapes of curves it can fit.
- Good baseline for most forecasting tasks.

Non-Linear Regression

- Can fit more complex types of curves.
- More effort required to find the best fit.
- Harder to interpret independent variables.

Bias-Variance Tradeoff

- Overfit (High Variance): Accurate on training set, but fails on test set.
- Underfit (High Bias): Inaccurate on both training and test sets.

Forecasting Models 1: Regression: Linear Vs Non-Linear? (/2)

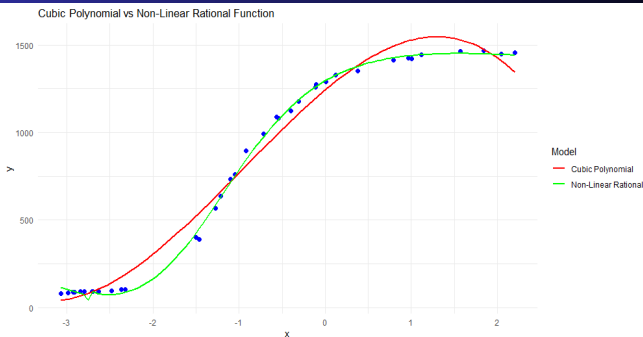


FIGURE 2.36: Electron Mobility in Electric Field: Linear & NonLinear Model

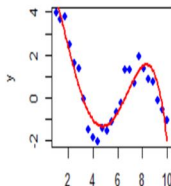
- **Linear** uses a cubic curve to fit the NIST Experimental data (Fig 2.36).
 - Fits data ok: high R^2 98.5%, Adj R^2 98.3%, In parts under/over-predicts $\Rightarrow$ bias.
- **Nonlinear** uses $y = (\beta_0 + \beta_1 x_t + \beta_2 x_t^2 + \beta_3 x_t^3) / (1 + \beta_4 x_t + \beta_5 x_t^2 + \beta_6 x_t^3)$
 - Curve has better fit due to unbiased nature and smaller residuals produced.
 - But not necessarily better results on validation/ test data.

Forecasting Models 1: Regression: LM vs. GLM

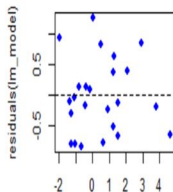
- Linear Model (LM): Form & Properties
 - Model Form: $y = X\beta + \epsilon$
 - Assumptions: $\epsilon \sim N(0, \sigma^2 I)$ (Homoscedasticity and Normality).
 - Estimation: Ordinary Least Squares (OLS) minimizes $\sum (y_i - \hat{y}_i)^2$.
 - Limitation: Poor fit for non-linear trends or non-constant variance.
- Generalized Linear Model (GLM): Purpose & Make-up
 - Extends LM to data beyond straight line or normal distribution.
 - Systematic: A linear combination of predictors ($X\beta$).
 - Link fn: Math'l "bridge": fits the linear predictors to the data shape.
 - Random: Allows for different types of "noise" or error patterns rather than just bell curves.
- Why use GLM over LM?
 - Flexibility: Models non-linear trends w/o "overfitting" (to noise).
 - Generalization: Yields more random residuals $\Rightarrow$ handles true signal not just memorizing points.

Forecasting Models 1: Regression: Linear Vs Non-Linear? (/3)

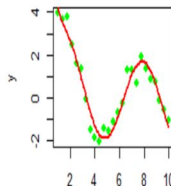
LM Fit (Polynomial Order 4)



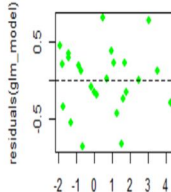
LM Residuals



Improved GLM Fit



GLM Residuals



- `lm()` residuals have patterns, $\Rightarrow$ overfitting.

- Polynomial `lm()` Model
 $R^2 = 0.878$, $\text{Adj } R^2 = 0.854$

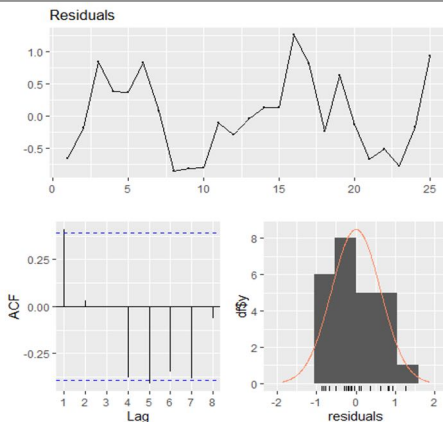
- Well-chosen nonlinear follows the data almost exactly.

- The residual plot displays the desired randomness from an unbiased model.

- `glm()` residuals are more random, $\rightarrow$ generalizes better.

- Nonlinear `glm()` Model
 $R^2 = 0.945$, $\text{Adj } R^2 = 0.937$

Forecasting Models 1: Regression: Linear Vs Non-Linear? (/4)



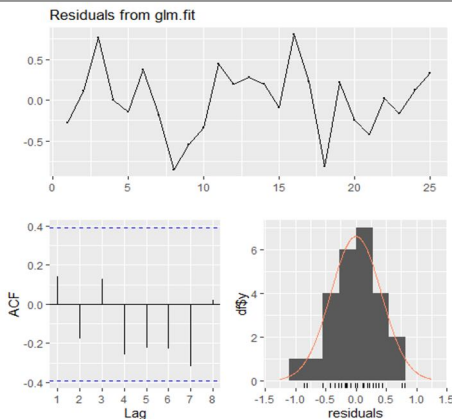
```
> checkresiduals(glm model)
```

```
data: Residuals
```

```
LM test = 18.017, df = 8, p-value = 0.0211
```

- To see detail, need to do more than eyeball the residuals fit.
- A Breusch-Godfrey test is like Ljung-Box only for regression residuals
- Examining the `lm()` residuals with this gives:
 - Significant lags in the ACF
 - Breusch-Godfrey: $p < 0.05$.
- $\Rightarrow$ conclude `lm()` has autocorrelation among the residuals.

Forecasting Models 1: Regression: Linear Vs Non-Linear? (/5)



```
> checkresiduals(glm_model)

data: Residuals from glm.fit
LM test = 15.273, df = 7, p-value = 0.03265
```

- How about the residuals from the `glm()` .?
- Examining the `glm()` residuals with this gives:
 - A slight pattern in residuals (top plot)
 - No significant lags in the ACF
 - A better p value but still $p < 0.05$.
- So `glm()` has still got autocorrelation among the residuals.
- Shows how difficult it can be to predict!

Forecasting Models 1: Regression: Exponential forms

- Easiest to fit a nonlinear model by transforming⁴ y &/or x before regression.
- Four varieties of these are shown in Table 2.4

$\begin{matrix} X \\ Y \end{matrix}$	X	$\log(X)$
Y	<i>linear</i> $y_i = \beta_0 + \beta_1 x_i$ $\beta_1 \text{unit}(Y) \leftarrow \text{unit}(X)$	<i>linear-log</i> $y_i = \beta_0 + \beta_1 \log x_i$ $\beta_1 \text{unit}(Y) \leftarrow \text{unit}(\ln X)$
$\log(Y)$	<i>log-linear</i> $\log y_i = \beta_0 + \beta_1 x_i$ $\beta_1 \text{unit}(\ln Y) \leftarrow \text{unit}(X)$	<i>log-log</i> $\log y_i = \beta_0 + \beta_1 \log x_i$ $\beta_1 \text{unit}(\ln Y) \leftarrow \text{unit}(\ln X)$

TABLE 2.4: Four Kinds of log Transform Model for Regression

⁴Usually with the natural log ($\ln$)

Forecasting Models 1: Regression: Exp'l forms (/2)

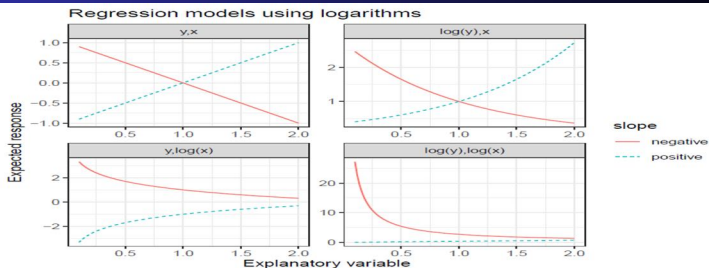


FIGURE 2.37: Four Kinds of log Transform Model for Regression

- 1 **linear-linear** (Top, L): unit increase in $X \rightarrow \beta_1$ units of Y
- 2 **linear-log** (Btm, L): unit increase in $\log X \rightarrow \beta_1$ units of Y .
— Or, β_1 is the change in Y when X is multiplied by e .
- 3 **log-linear** (Top, R): unit increase in $X \rightarrow$ increase in $\log Y$ of β_1 units.
— This means that Y is multiplied by e^{β_1} .
- 4 **log-log** (Btm, R): unit increase in $\log X \rightarrow \beta_1$ units increase in $\log Y$.
— β_0 : resembles an *elasticity* (i.e. bulk response to a force)
— β_1 : average percent change in Y from percent increase in X

Forecasting Models 1: Regression: Different Types

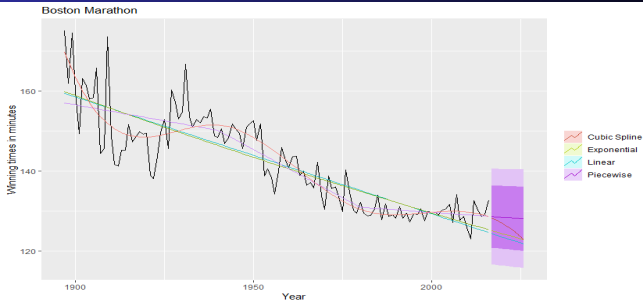


FIGURE 2.38: Boston marathon winning times in minutes (1897-2015)

- In Fig 2.38 trend, volatility decline with better times, less variation.
- Log-Linear Model used here as exponential model $\ln y_t = \beta_0 + \beta_1 t + \epsilon$
- 3 sub-periods so a piecewise linear model: (a) Up to 1940 (much volatility), (b) Post-1940 (linear decrease), (c) post-1980 (flattens out)
- Cubic Spline model gives better forecasting than piecewise linear?

Forecasting Models 2a: Moving Average Smoothing

Year	Qtr	Dec Value	Sold	4pt MA	Trend	DeTrend	Seasonal MA	Corrected Seasonal MA	Seasonally Adjusted	forecast	Seasonal MA	
2014	1	2014.25	22								-6.83333	-6.76042 Q1
	2	2014.5	25								-5.08333	-5.01042 Q2
	3	2014.75	44	30.25	30.625	13.375	13.33333333	13.40625	30.59375		Sum of Mas	-0.29167 13.3333 13.4063 Q3
	4	2015	30	31	31.125	-1.125	-1.708333333	-1.635416667	31.6354167		Correction/Q	-0.07292 -1.70833 -1.63542 Q4
2015	1	2015.25	25	31.25	31.5	-6.5	-6.833333333	-6.760416667	31.7604167		Must sum to 1	
	2	2015.5	26	31.75	31.875	-5.875	-5.083333333	-5.010416667	31.0104167	Trend		
	3	2015.75	46	32	32.375	13.625	13.33333333	13.40625	32.59375	Last	39.125	
	4	2016	31	32.75	33.375	-2.375	-1.708333333	-1.635416667	32.6354167	First	30.625	
2016	1	2016.25	28	34	34.375	-6.375	-6.833333333	-6.760416667	34.7604167	Change	8.5	
	2	2016.5	31	34.75	35.25	-4.25	-5.083333333	-5.010416667	36.0104167	Ave Chang	0.77272727	
	3	2016.75	49	35.75	36	13	13.33333333	13.40625	35.59375			
	4	2017	35	36.25	36.625	-1.625	-1.708333333	-1.635416667	36.6354167			
2017	1	2017.25	30	37	37.625	-7.625	-6.833333333	-6.760416667	36.7604167			
	2	2017.5	34	38.25	39.125	-5.125	-5.083333333	-5.010416667	39.0104167			
	3	2017.75	54	40	39.8977							
	4	2018	42		40.6705							
2018	1	2018.25	35		41.4432							
	2	2018.5	37		42.2159							
	3	2018.75	56		42.9886							
	4	2019	42		43.7614							

2. Next need to take the Seasonal MA, divide by sum of Q1 to Q4 to normalize the SMA values.

1. Trend is projected Forward first. Need to calculate 2018 Qtr3 & 4 values. Add 0.772 to last.

3. The final step is to add the projected trend values to the corrected Seasonal MAs.

FIGURE 2.39: Quarterly Sales 2012-19 (Red is Forecast)

- Saw above additive decomposition for time series modelling.
- See in Fig 2.39 we just use same method for forecasting
- Forecast sales are: Projected Trend first added to Seasonal MA

Forecasting Models 2a: Moving Average Smoothing (/2)

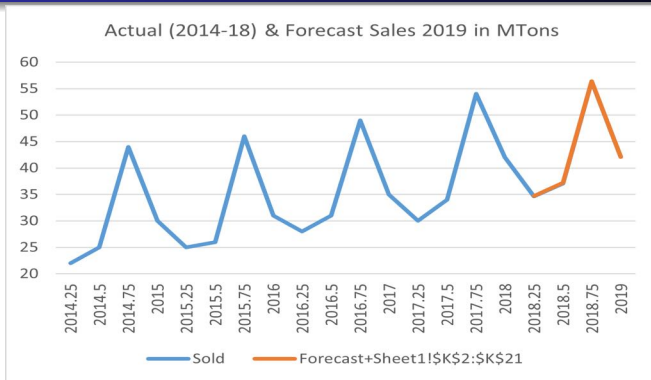


FIGURE 2.40: Quarterly Sales 2012-19 (2019 is Forecast)

- Fig 2.40 shows the resulting quarterly graph & forecast
- The forecast method is reasonably accurate as it uses forecast trend & quarterly MAs

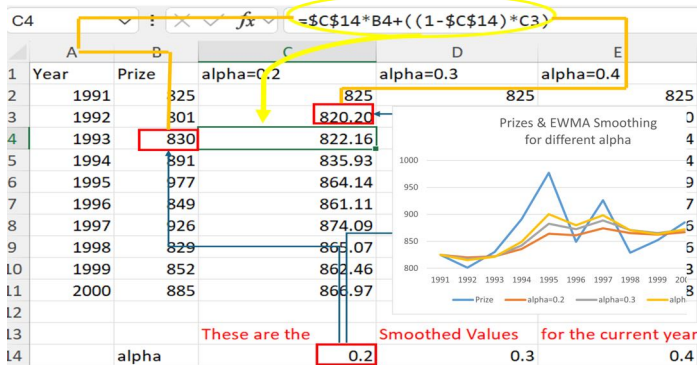
Forecasting Models 2b: EWMA Methods

Recall Exponentially Weighted Moving Average

- 1. Smoothing (Current Average)
 - Goal: Level out the data values to see the current year's trend.
 - Excel Logic: Uses the *value* from the same row.
 - Formula: $EM A_t = \alpha x_t + (1 - \alpha)EM A_{t-1}$
 - Visual: The line follows the peaks and valleys without a delay.
- 2. Forecasting (Future Guess)
 - Goal: Predict next year's value using today's average.
 - Excel Logic: Uses the *value* from the row above (the past).
 - Formula: $\text{Forecast}_{t+1} = EM A_t$
 - Visual: The line looks shifted one year to the right.
- Key Difference
 - Smoothing uses "today" to explain "today."
 - Forecasting uses "today" to guess "tomorrow."

Forecasting Models 2b: EWMA Methods: Smoothing

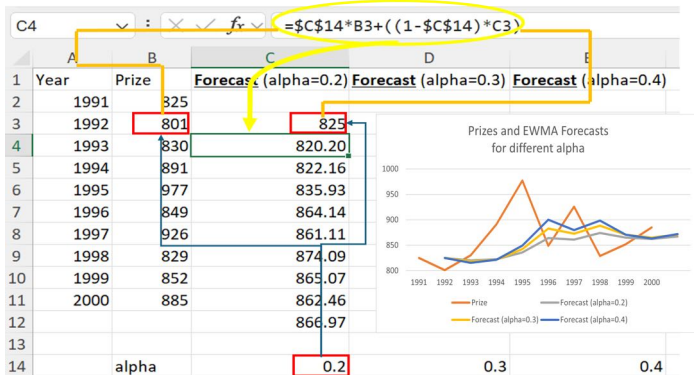
- Goal: Calculate the average for the year we are actually in.
- Excel Logic: Uses the Prize from the **current** row (B4).
- Formula: $EMA_t = \alpha x_t + (1 - \alpha)EMA_{t-1}$



- Inset: The line follows the prize peaks without a delay.

Forecasting Models 2b: EWMA Methods: Forecasting

- Goal: Predict next year's prize using today's average.
- Excel Logic: Uses the Prize from the **row above** (B3).
- Formula: $Forecast_t = \alpha \times B3 + (1 - \alpha) \times Forecast_{t-1}$



- Visual: The line looks shifted one year to the right.

Forecasting Models 3a: ARIMA Models

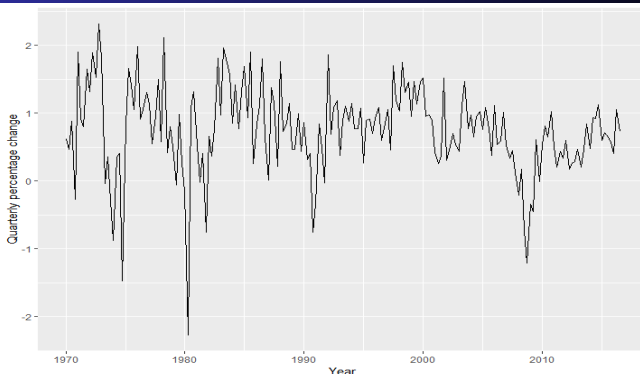


FIGURE 2.41: Qly % change in US consumption Exp (from Hyndman)

- Saw above how ARIMA models can be fitted in general.
- No seasonality apparent $\Rightarrow$ non-seasonal ARIMA model is ok for data.
- In Fig 2.41 use `auto.arima()` for p, d, q in quarterly ARIMA model of US Consumption Expenditure.

Forecasting Models 3a: ARIMA Models (/2)

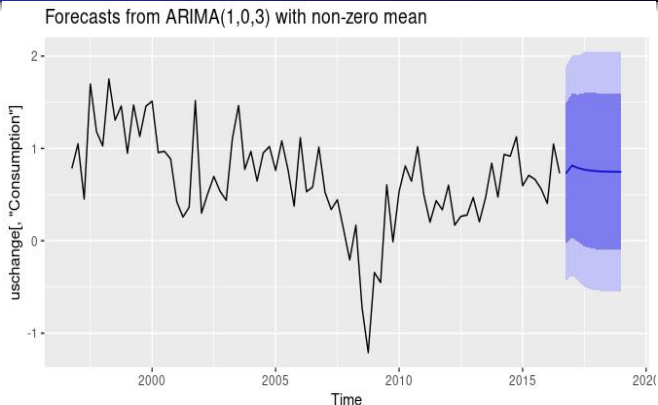


FIGURE 2.42: Forecast Model of US Consumption (from Hyndman)

- The ARIMA model in Fig 2.2.42 fitting the data in Fig 2.2.41 is:

$$x_t = c + 0.589x_{t-1} - 0.353w_{t-1} + 0.084w_{t-2} + 0.174w_{t-3} + \epsilon_t$$

Forecasting Models 3b: SARIMA: Eurretail Data

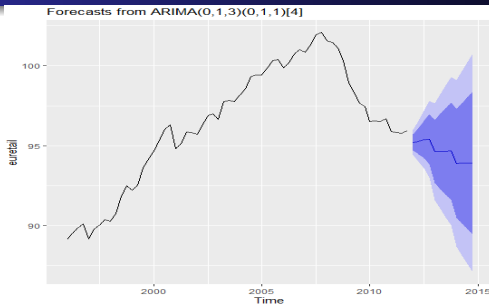


FIGURE 2.43: Eurretail Forecast with 80% and 95% prediction intervals

- Fig. 2.43 has Eurretail forecast with fitted $ARIMA(0, 1, 3)(0, 1, 1)_4$ model
- Coeffs: $MA1 = 0.263$, $MA2 = 0.369$, $MA3 = 0.420$, $SMA1 = -0.664$
- Forecasts:
 - follow recent trend in the data, due to double differencing.
 - increasing prediction intervals show increase/ decrease possible
 - point forecasts trend downwards; prediction intervals allow for data to trend upwards during the forecast period.

Forecasting Models 3b: SARIMA: Steroid Data

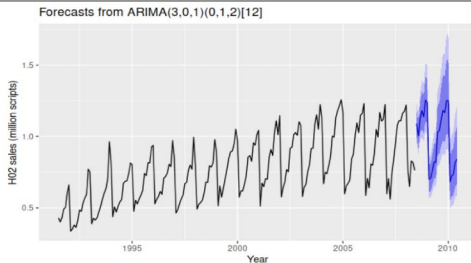


FIGURE 2.44: Steroid Data Forecast with 80% and 95% prediction intervals

- Fig. 2.44 has Corticosteroid forecast with fitted $ARIMA(3, 0, 1)(0, 1, 2)_{12}$
- Coeffs: $AR1 = -0.160$, $AR2 = 0.548$, $AR3 = 0.568$, $MA1 = 0.383$, $SMA1 = -0.522$, $SMA2 = -0.177$
- Some spikes in the ACF, and the model fails the Ljung-Box test.
- Can still use model for forecasting, but prediction intervals may not be accurate due to correlated residuals.

Time Series Forecasting: Session Learning Outcomes

In this session on forecasting, we have:

- ① Learned about Qualitative & Quantitative forecasting
- ② Seen the difference between *ex post* and *ex ante* forecasting
- ③ Looked at overfitting & underfitting and the tradeoff between Bias & Variance
- ④ Compared different models for Forecasting in Time Series Analysis:
 - + Naïve models
 - + Linear & Non-Linear Models
 - + Moving Average Smoothing Models
 - + ARIMA & SARIMA Models
- ⑤ Seen how the above perform for different examples

Summary

In this section on Time Series, we covered many basic aspects of Time Series:

- What characterises different forms of time series & how they arise
- Outlined the main **objectives** of time series analysis
- Inferred key **aspects** arising out of studying time series
- Described the **trend** in the series using Moving Averages & Linear Regression
- Estimated the **Key Patterns** (Trend/ Seasonal/ Cyclical effects) in a series & removing them.
- Learned how to construct non-seasonal & seasonal ARIMA models
- **Identified** & **interpreted** future values using first-order autoregressive & moving average models.